ASSIGNMENT 3

Rainfall-Aware Landslide Risk Mapping for Himachal Pradesh

Interim progress: from a risk map to a daily landslide early-warning system for HP's disaster authorities (problem, methodology, implementation, results)

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Slide 3 recaps Presentations 1 and 2; slides 5–6 list what changed since then.

SUMMARY

The project at a glance

We built and tested the core of a daily warning system; it separates storm days from quiet days, and we know its limits

Problem

- Rain-triggered landslides block roads and kill people in Himachal Pradesh (HP) every monsoon
- Published HP studies give static maps (where), not daily warnings (when)
- Goal: a daily, district-level warning for HP's disaster authorities, using free data only

What we built

- 981 dated landslides (HP, Uttarakhand, J&K) with 42 features each
- 2,167 hard 'no-landslide' examples (training table: 3,136 rows)
- A model tested honestly (whole years hidden) and a whole-HP daily scoring of 55,253 cells of 1 km

Honest limits

- Catches about half of recorded HP slides at the lower alert level
- Rain alone times slides as well as the model: the satellite rain data sets the ceiling
- Records are biased to roads; remote areas are less certain

0.775

Live model skill (ROC AUC), whole years hidden; 0.5 = coin toss

0.856

Same model with the easy random-row test: why a high score can mislead

66.8%

of HP flagged on a storm day (14 Aug 2023), all 31 recorded slides inside

0.4%

of HP flagged on a quiet day (12 Sep 2023), 0 of 12 districts alerted

Next: daily automation and email alerts, storm replays to set the final alert rules, impact-assessment design, final paper.

Source: Our analysis of the landslide inventory (GSI Bhusanket, NASA Global Landslide Catalog) with NASA GPM IMERG rainfall; tests on slide 12; maps on slides 21–22.

RECAP

Where the project stood after Presentations 1 and 2

We proposed rainfall-aware risk mapping, built a first dataset and model, and were told our negative examples were the weak point

Presentation 1 (17 Aug): proposal

- Idea: risk = terrain susceptibility × rainfall, for HP, delivered to disaster authorities (not the public)
- AHP-weighted terrain map as a baseline
- Research questions: RQ1 does rainfall improve on terrain alone; RQ2 new vs reactivated slides; RQ3 terrain–land-use–rain combinations
- Feedback: AHP maps are common; rainfall-based prediction is under-explored but needs dated landslides; small road slides are too small for open satellite data

Presentation 2 (17 Sep): first build

- GSI's national inventory extracted from a 904-page PDF (36,071 records); only 1.7% of HP records have an exact date
- Uttarakhand and J&K added by a two-stage method; 886 dated events
- CHIRPS rainfall for 3–30 days before each event
- First models on slope + rain with random background points: Random Forest AUC 0.895

Feedback on Presentation 2

- 'Good effort in assembling a dataset'
- 'Negative sampling has some serious issue': a slope-and-rain model gives high recall but low precision, so many false alarms and users stop trusting it
- This feedback drove most of the work since (slides 5–6)

Source: Our Presentation 1 (17 Aug 2026) and Presentation 2 (17 Sep 2026) decks and the instructor's written feedback.

PROBLEM

Why Himachal Pradesh needs a daily landslide warning

The data to warn exist every day for free; what is missing is a tested system that turns them into district alerts

The gap

- Monsoon rain triggers landslides across HP every year; in 2023 they caused hundreds of deaths and closed hundreds of roads
- Published HP studies (Kullu valley, AUC 0.79–0.91; Beas basin, AUC 0.86) produce static maps: where, not when
- GSI's regional landslide forecasting covered only Shimla and Kinnaur in HP as of the 2025 monsoon

Our goal and user

- Every morning, score landslide risk for every 1 km cell of HP
- Give each of the 12 districts a level: no alert, Watch or Warning
- Email the district and state disaster authorities; show the map in a web app
- Built on free global data, so a partner organisation can run it

195

dated HP landslides (2005–2026) in our cleaned inventory

12 districts

each to receive a daily level: none, Watch or Warning

55,253 cells

of 1 km covering the whole state, scored every day

Source: Meena et al. 2019; Singh et al. 2024; Ministry of Earth Sciences, Lok Sabha reply 2025 (GSI coverage); 2023 toll: state figures in the press, Aug 2023.

What we changed since Presentation 2

Each point of feedback was investigated, measured and fixed, not just acknowledged

Feedback or issue	What we found when we checked	What we changed (Oct–Nov)
Negative sampling has a serious issue	Old negatives were random: median 3,868 m high (landslides: 1,228 m), 408 of 878 above 4,000 m, km away from roads	2,167 hard negatives of two kinds: same place on a safe day, and similar slopes on the storm day
Score looked high (RF AUC 0.895)	Easy negatives plus random-row testing inflate the score: the same model scores 0.856 with random rows but 0.775 with whole years hidden	All testing now hides whole years; we report the strict number
Model used only slope and rain	Too few factors to tell stable wet slopes from failing ones	42 features in 9 groups (terrain, drainage, roads, rock, faults, landforms, land cover, vegetation, rain/soil/snow)
Event dates (our own audit)	Some dates were wrong; 2 coordinates had typos; 5 duplicates	Inventory re-audited: 981 cleaned, dated events
AHP maps are common (Presentation 1)	Agreed with the literature	AHP kept only as a baseline; contribution is daily, rain-aware prediction

Net effect

A lower but honest score (0.775 instead of 0.856–0.895), from a model whose behaviour we can check: what it learned, where it fails, and why

Source: Our analysis of the 878 background points from Presentation 2 and of the re-audited inventory; testing comparison on slide 12.

Numbers that changed since Presentation 2, and why

Every figure in this deck that differs from Presentation 2 is listed here with the reason; one earlier description was wrong

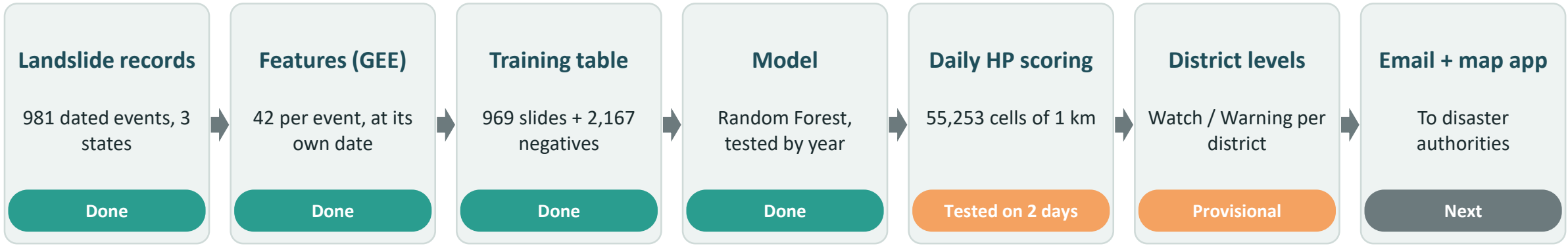
Item	Presentation 2	Now	Why it changed
Dated landslide events	886	981	A corrected date parser recovered more exact dates; the audit then removed 5 duplicates and 1 undatable event
No-landslide examples	878 random points	2,167 hard negatives	Instructor's feedback (slides 10–11)
Features	5 (slope + 4 rain windows)	42 (38 used by the model)	Full set of terrain, geology, land-cover and weather factors
Best score (ROC AUC)	0.895 (random rows)	0.801 full / 0.775 live (whole years hidden)	Harder negatives and stricter testing (slide 12)
Most important factor	30-day rain	Rain on the day; road distance	The old negatives were dry mountain tops, so long-term rain looked decisive (slide 20)
Where the 878 background points were	'Inside HP, 500 m buffer'	**Spread over HP, UK, J&K and beyond**	Our earlier description was wrong; corrected here
Rain 'trigger-consistency' check	23.6% vs 11.7%	Dropped	It used the wrongly parsed dates; with correct dates it reverses
AHP terrain baseline	6-factor map	Kept as baseline	Not yet scored on the same test as the new model (final report)

Source: Our Presentation 2 deck; our re-audit of the inventory and of the Presentation 2 background points.

APPROACH

System overview: from records to daily district alerts

The full chain is built up to a whole-HP daily map; the automatic daily run and email are next



Data used (all free)

- Rain: NASA GPM IMERG (half-hourly), CHIRPS; forecast: NOAA GFS
- Terrain: Copernicus DEM 30 m, MERIT Hydro; land: ESA WorldCover, Landsat
- Geology: GSI National Geoscience Data Repository; roads: OpenStreetMap

Tools

- Google Earth Engine (Python API) for all geodata, on Google's servers
- Python (scikit-learn) for models; every step scripted and logged
- Planned: GitHub Actions for the daily run, email alerts, an Earth Engine map app

Source: Data providers: NASA (GPM IMERG), UC Santa Barbara (CHIRPS), NOAA (GFS), ECMWF (ERA5-Land), ESA/Copernicus, USGS (Landsat), GSI, OpenStreetMap; computed in Google Earth Engine.

DATA

Building a dated landslide inventory

A warning model needs landslides with exact dates; only 1.7% of HP's national records had one, so we pooled two sources and two similar states

904-page PDF
GSI Bhusanket national inventory

36,071 records
extracted; HP 6,607

HP: 114 with a day-level date
only 1.7%; most give month and year only

+ NASA Global Landslide Catalog
all dated; filtered by location accuracy

981 cleaned events
HP 195 · Uttarakhand 548 · J&K 238

Why add Uttarakhand and J&K

- HP alone has too few dated slides to learn from
- Chosen by a method, not by eye: (1) same Himalayan thrust system as HP; (2) similar share of rain-triggered slides
- Ladakh (12 points, cold desert) is left out of training

Cleaning (all logged)

- Dates parsed to the day; impossible dates removed
- 2 coordinate typos fixed against Survey of India sheets; 1 wrong year fixed; 5 duplicates removed
- 242 of 981 events fall on one day (16 Jun 2013), so testing must hide whole storms or years

Source: GSI Bhusanket inventory (bhusanket.gsi.gov.in); NASA Global Landslide Catalog (Kirschbaum et al. 2015); Survey of India sheet grid for coordinate checks.

DATA

42 features per event, computed at each event's own date

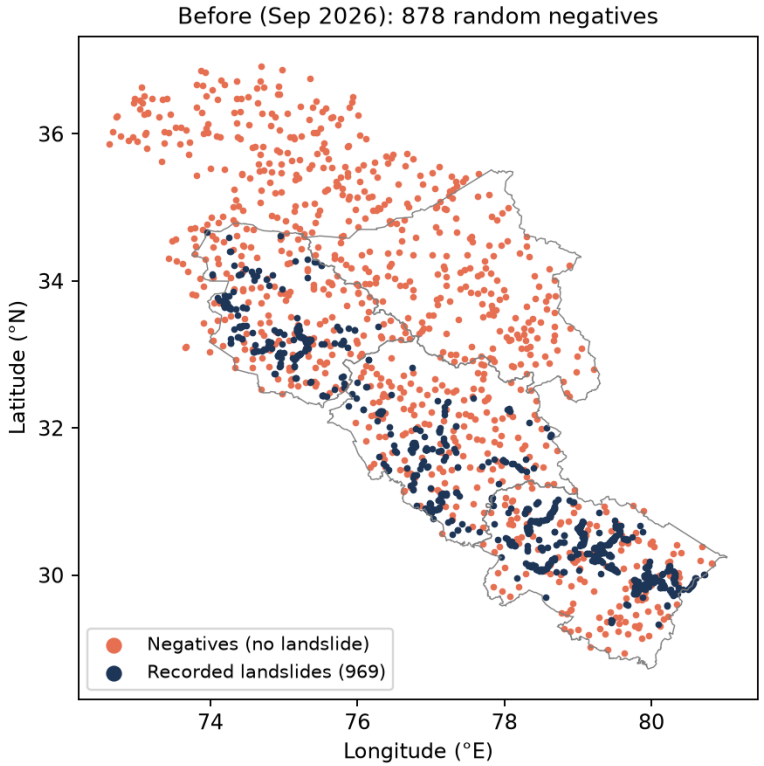
Each feature group has a published basis or a data test, and the same code serves training and the daily map

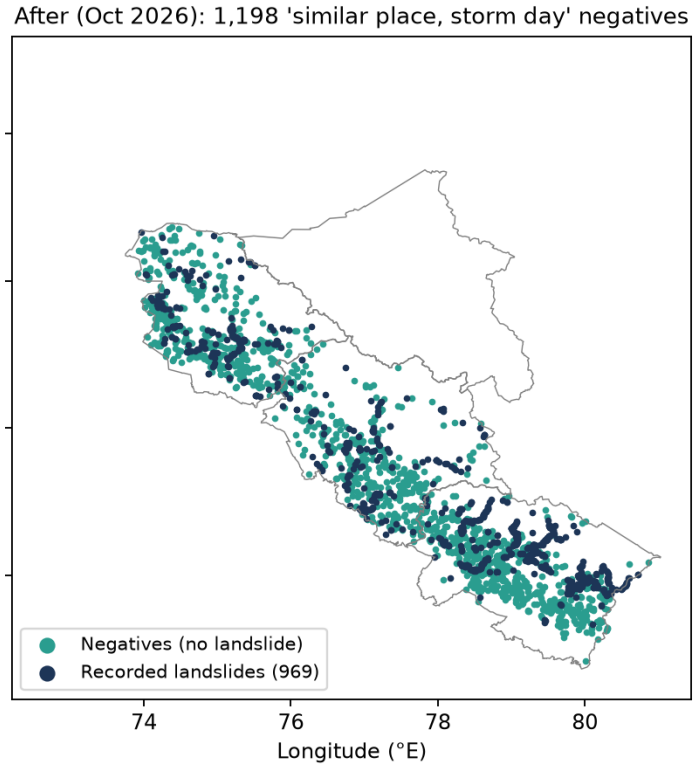
Group (count)	Features	Source	Why it matters
Terrain (9)	elevation, slope, aspect, curvature, position, wetness index	Copernicus DEM 30 m, MERIT Hydro	Core landslide factors (Reichenbach et al. 2018)
Drainage (2)	distance to stream, height above stream	MERIT Hydro	Toe erosion, saturation
Roads (2)	distance to any road, to a major road	OpenStreetMap	Road cuts destabilise slopes (Froude and Petley 2018)
Rock (2)	rock class, main rock	GSI geology 1:50k	Weak vs strong rock
Faults (2)	distance to fault, to thrust	GSI geology 1:50k	Fractured rock near faults
Landforms (1)	landform class	GSI geomorphology 1:250k	Dissected hills fail more
Land cover (2)	class at point, local majority	ESA WorldCover 2021	Vegetation, built-up
Vegetation (2)	NDVI over the 365 days before	Landsat 5/7/8/9	Root strength; no post-event leakage
Rain, soil, snow (18)	rain on the day and 1–30 days before; soil water; snow	GPM IMERG, CHIRPS, ERA5-Land	The daily trigger (as in NASA LHASA v2)
Data flags (2)	rain-data completeness	IMERG	Removed before modelling

Source: Reichenbach et al. 2018; Froude and Petley 2018; Stanley et al. 2021 (LHASA v2); data providers on slide 7.

Why the old 'no-landslide' examples were wrong

Random negatives sat on high, icy, remote ground, so a model could score well without learning anything about rain





3,868 m

median height of old negatives (landslides: 1,228 m)

408 of 878

old negatives above 4,000 m: snow, glaciers, bare rock

4.8 km

median distance to a road (landslides: 22 m)

Blue: recorded landslides. Left: old random negatives. Right: new 'similar place, storm day' negatives.

Source: Our analysis of the 878 background points used in Presentation 2 and of the new negatives (elevation: Copernicus DEM; roads: OpenStreetMap).

New design: two kinds of 'hard' negatives

One kind teaches WHEN (same place, safe day), the other teaches WHERE (similar slope, storm day)

Type 1: same place, safe day (969)

- Exactly the landslide's location, on a different day
- No recorded slide within 5 km and 30 days of that day
- Months follow the landslides' mix (88% June–September), so many safe days are rainy too
- Only the weather differs, so this measures how well the model reads rain

Type 2: similar place, storm day (1,198)

- A random spot in the same state on the landslide's date
- Slope at least 10.9° and within 284 m of a motorable road (like 90% of our slides)
- At least 2 km from any recorded slide; at most 10 per storm day
- The storm is shared, so this teaches which ground fails

Median rain (IMERG, mm)	On the day	7 days before	30 days before
Recorded landslides	34	89	217
Type 1: same place, safe day	1.5	30	145
Type 2: similar place, storm day	8	65	269

Training table: 969 landslides + 2,167 negatives = 3,136 rows. All rules re-checked: 0 violations.

Source: Steger et al. 2024, Geoscience Frontiers 15(3):101822; Zhu et al. 2024, Geomatics, Natural Hazards and Risk 15(1); thresholds from our own landslide data.

Honest testing: hide whole years, not random rows

Rows from the same storm share weather; testing on random rows lets the model 'see' the storm it is scored on

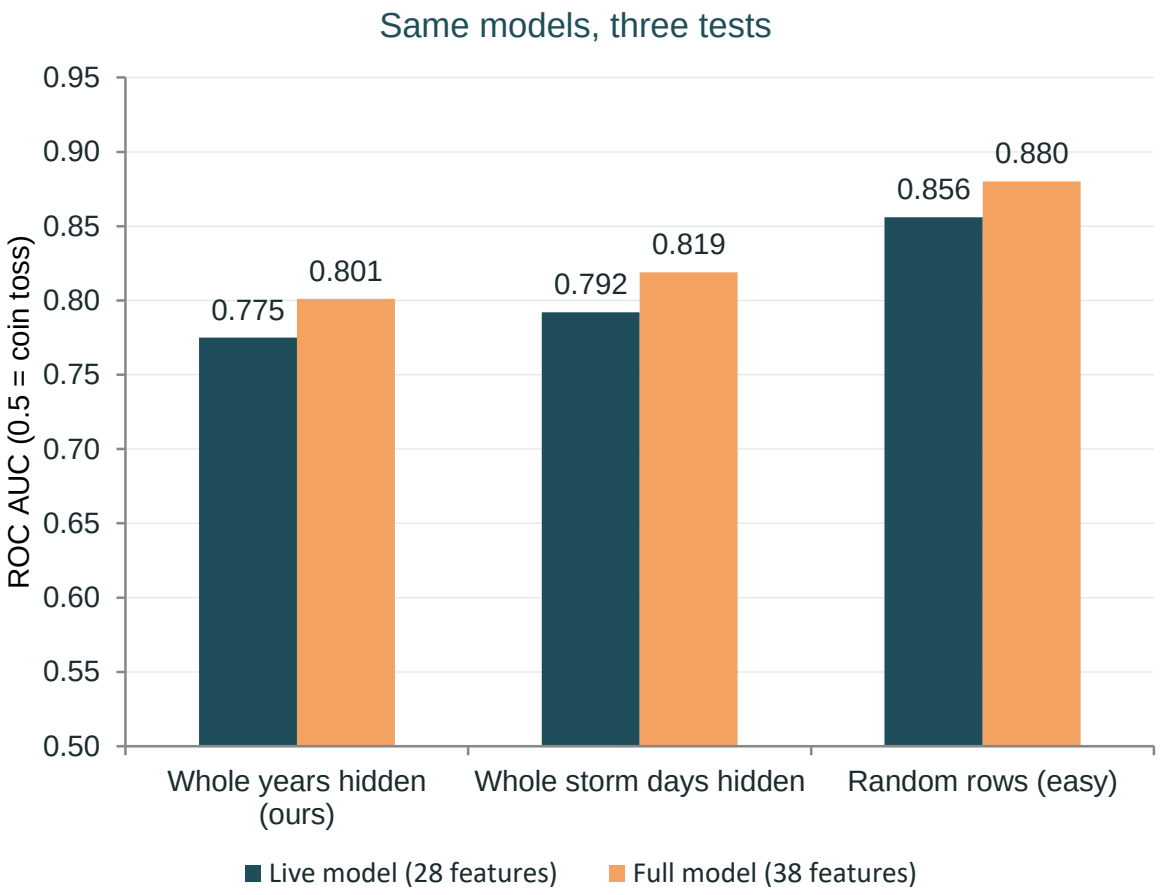
How it works (simplified):

2010	2011	2013	2015	2016	2023	2025

Grey years: train the model. Orange year: scored by that model, which never saw any storm of that year. Repeat until every year has been scored once; then compute one score over all rows.

What the chart shows

- Same model, three ways of testing
- Random rows: 0.856 (live) and 0.880 (full model)
- Whole years hidden: 0.775 and 0.801, the numbers we report
- The gap (about 0.08) is the optimism of the easy test



Source: Roberts et al. 2017, Ecography 40(8):913–929; Brenning 2005, NHESS 5:853–862; our cross-validation results (whole years held out).

METHOD: MODEL CHOICE

Three model types, compared fairly

XGBoost scored highest but leaned on roads instead of rain; we chose Random Forest

Model (38 features, whole years hidden)	ROC AUC, all	ROC AUC, HP	Without road features	Weather-only test	Verdict
Logistic regression	0.780	0.753	0.762	0.744	Baseline
Random Forest	**0.801**	**0.790**	0.755	**0.746**	**Chosen**
XGBoost	0.807	0.804	0.737	**0.722**	Clearly worse on weather-only test

The weather-only test

- A daily warning lives or dies on reading the weather
- Landslide days vs safe days at the same places: roads, rock and slope are identical, only the weather differs
- The rule for 'clearly worse' was fixed before the test

Why Random Forest

- Almost as good overall as XGBoost (0.801 vs 0.807)
- Not worse than road-free models at reading the weather; XGBoost was
- Fewer false alarms on HP safe days at the same catch rate

Source: Breiman 2001 (Random Forest); Chen and Guestrin 2016 (XGBoost); Stanley et al. 2021 (LHASA v2); our cross-validation results (whole years held out).

The road bias in landslide records, and how we correct for it

Slides are recorded where they hit roads; we keep road distance in training but score every cell as if it were watched like a roadside slope

22 m

median distance of a recorded landslide to a road

1st / 2nd

rank of 'distance to major road' among 38 features in every model

70 (9 in HP)

recorded slides more than 500 m from a road: few to test remote skill with

Live model; same safe-day false alarms (16%)	Real road distances	Road-neutral
All recorded slides caught	37%	49%
Remote slides caught (> 500 m from road)	29%	36%
Weather-only test (ROC AUC)	0.683	0.741
Other slopes flagged on storm days	3%	18%

Decision and its cost

- Road-neutral scores drive the whole-HP map and the district alerts; real-road scores kept as a second layer
- Gain: more slides caught, including remote ones, at the same false-alarm rate on safe days
- Cost: on storm days much more of the map is flagged; remote false alarms cannot be measured

Source: Steger et al. 2017, Landslides 14:1767–1781; Steger et al. 2021, Sci. Total Environ. 776:145935; our cross-validation results (whole years held out).

METHOD: LIVE DATA

Live model: only data available on the morning it runs

Two rain and soil sources arrive weeks late, so the daily system uses a slightly weaker 'live' model

Data	Delay in Earth Engine	Usable every morning?
NASA GPM IMERG rain	about 2 days	**Yes**
NOAA GFS forecast rain	runs complete in 6–12 h	**Yes (fills the last 2 days, today, tomorrow)**
ERA5-Land soil moisture, snow	about 12 days	**No**
CHIRPS rain	about 24 days (monthly)	**No**

Whole years hidden	Full model (38)	Live model (28)
ROC AUC, all states	0.801	0.775
ROC AUC, HP	0.790	0.763
Weather-only test	0.746	0.683
HP safe days flagged at Watch	16%	26%
HP safe days flagged at Warning	5%	13%

What this means

- The full model is the best possible with complete data; the live model is what can actually run each day. We report both.
- The cost of the delays: the same share of slides is caught, with more false alarms (26% vs 16% of safe days at Watch).
- Rejected: putting IMERG values into the CHIRPS slots; in these mountains the two correlate only 0.53–0.75.

Source: Data delays checked in the Google Earth Engine catalogue (Oct 2026); our cross-validation results (whole years held out).

Two alert levels, set provisionally

Watch and Warning trade missed slides against false alarms; final values will be set from storm replays with rules fixed in advance

Level (HP, chosen model)	Score	HP slides caught	HP safe days flagged	About false alarms per month at one spot
Watch: be ready	≥ 0.35	55%	16%	4.8
Warning: act	≥ 0.45	33%	5%	1.5

Why provisional

- Levels are set per 1 km cell, but alerts go out per district
- The district rule (now: a district is alerted when 5% of its cells are) proved far too loose on the storm day
- Both are fixed together after replaying several storms and quiet periods

Rules for the replays, written before running them

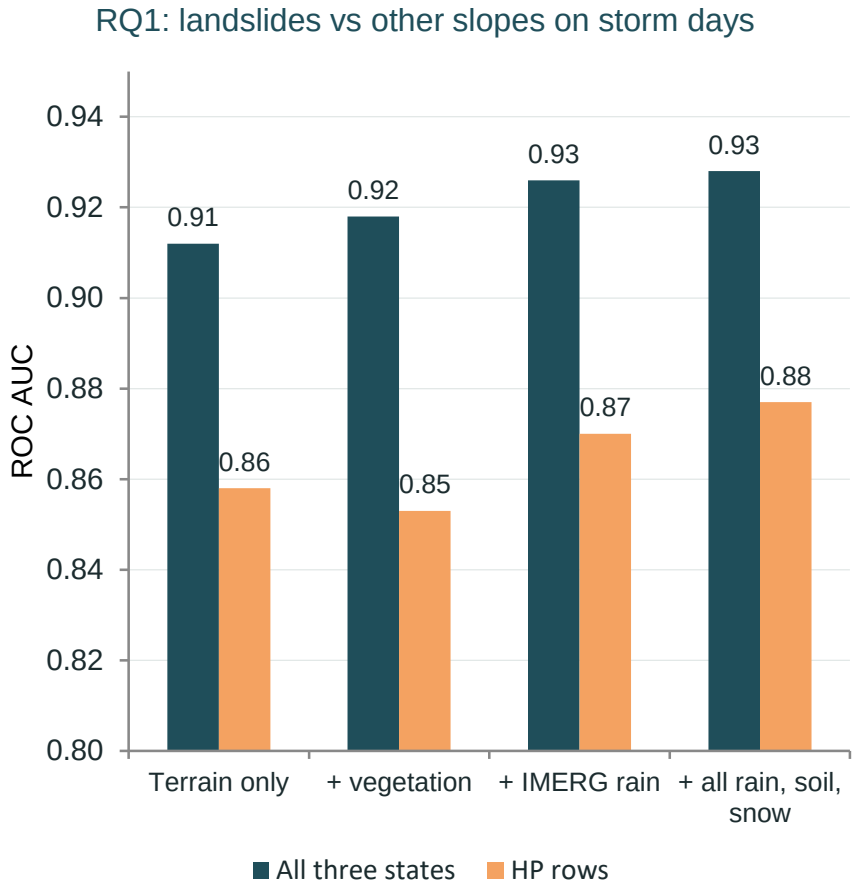
- Hit: the district is at Watch or higher on the event day or the day before
- Quiet days: count alert days on monsoon days with no recorded slide
- Each storm scored by a model that never saw its year
- Context: NASA's LHASA detected 8–60% of catalogued landslides

Source: NASA GES DISC, Global Landslide Nowcast (LHASA) README; Kirschbaum and Stanley 2018, Earth's Future 6:505–523; our cross-validation results (whole years held out).

Our three research questions: where they stand

RQ1 is answered: rainfall is what makes daily timing possible, and adds a little to locating slides; RQ2 and RQ3 are not started

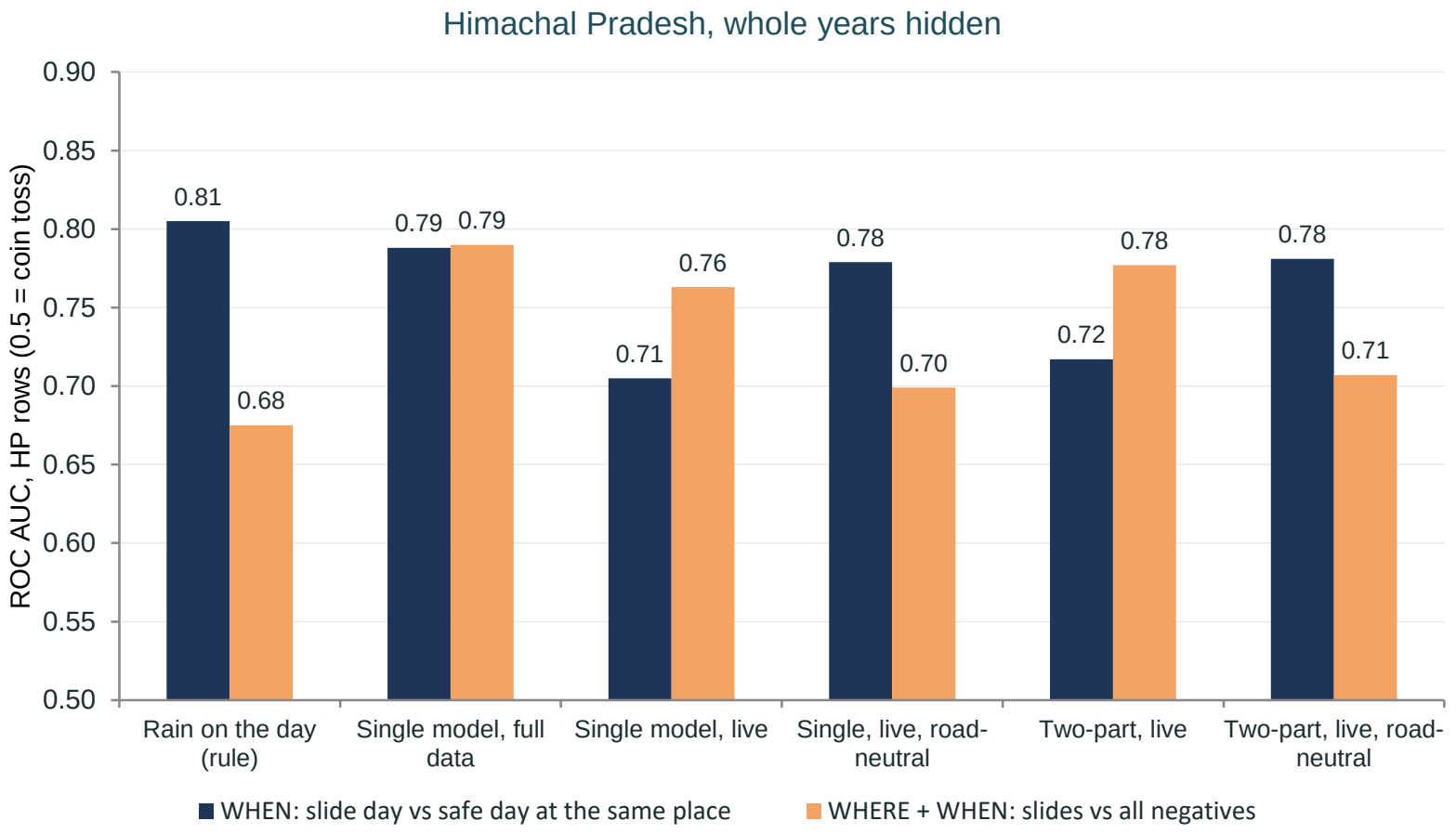
Research question (Presentation 1)	Status	Answer so far
RQ1: Does past and forecast rain improve prediction beyond terrain alone?	**Answered (past rain)**	WHEN a slide happens can only come from rain (terrain is the same every day). For WHERE on a storm day, rain adds a little on top of terrain (chart). Forecast rain: live system only, to be measured in the replays
RQ2: Can activity records tell new slides from reactivated ones?	**Not started**	Needs the inventory's free-text movement and activity fields cleaned first
RQ3: Which terrain, land-use change and rain combinations go with persistent movement?	**Not started**	Land-cover change needs more than the two WorldCover years (2020, 2021) available



Source: Research questions as set in our Presentation 1; our cross-validation results (whole years held out).

Does the model beat a simple rain rule?

Rain alone tells WHEN a slide happens about as well as any model; the model adds WHERE



What we learn

- WHEN: rain on the day alone (0.805) is as good as or better than every model
- WHERE + WHEN: every model beats rain alone (up to 0.79 vs 0.68)
- Why: half our negatives are other slopes on the same storm days, as wet as the slides, so one model learns to weigh rain less
- No model can time slides better than its rain input; satellite rain pixels are about 10 km wide

Source: IMD national bulletins (heavy rain: 64.5–115.5 mm per day); our cross-validation results (whole years held out).

A two-part model: WHERE × WHEN

Splitting the questions, as NASA's LHASA does, improves the overall score and halves wrong flags on other slopes

WHERE

Which slopes fail?

Landslides vs other slopes on the same storm day; terrain, rock, roads, vegetation

×

WHEN

Which days are dangerous?

Landslides vs safe days at the same place; rain only

= daily risk score

Live data, whole years hidden	Single model	Two-part model
ROC AUC, all states	0.775	0.813
ROC AUC, HP	0.763	0.777
HP safe days flagged (55% caught)	25.6%	21.5%
Other slopes flagged on storm days	13.9%	7.4%
Timing (HP); rain alone 0.805	0.705	0.717

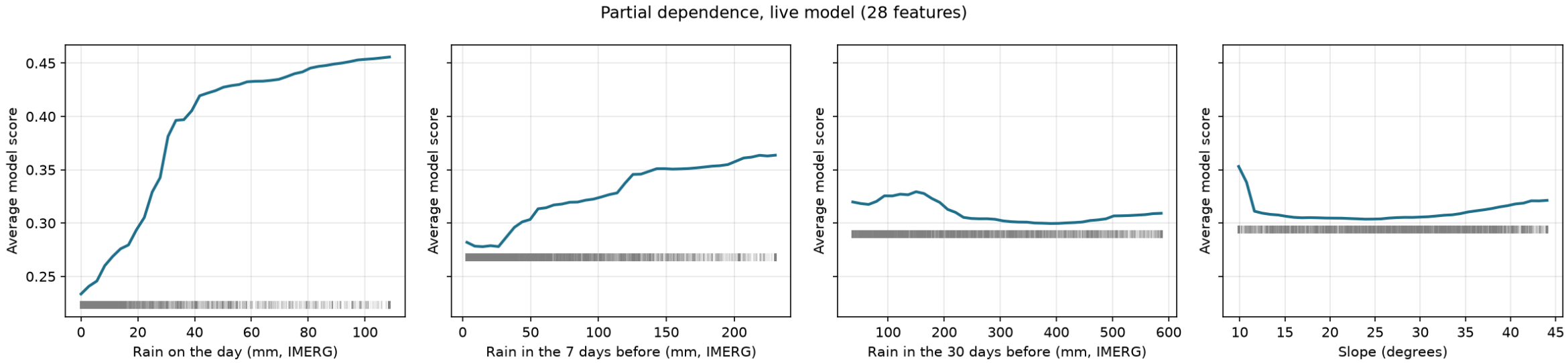
Reading

- Better overall and fewer wrong flags on other slopes: a real improvement, and a structure used by an operational system
- Timing improves only slightly: the satellite rain data, not the algorithm, limits how well slides can be timed
- Status: candidate for the final live model; to be confirmed in the storm replays

Source: Kirschbaum and Stanley 2018, Earth's Future 6:505–523; our cross-validation results (whole years held out).

Physics check: what the model learned

Rain effects look physical; two odd curves trace back to how the negatives were sampled



Rain on the day

Score doubles (0.23 to 0.46), steepest between 10 and 40 mm, then levels off

Physical

Rain in the 7 days before

Steady rise (0.28 to 0.36): wet ground fails more easily

Physical

Rain in the 30 days before

Flat: storm-day negatives are just as wet over 30 days

Sampling artefact

Slope

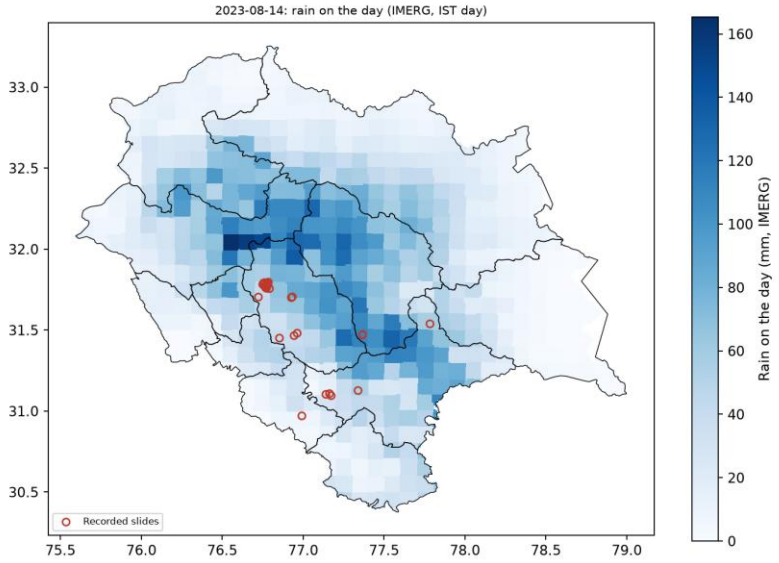
Spike near 10° (negatives only drawn on slopes ≥ 10.9°); gentle rise above 30°

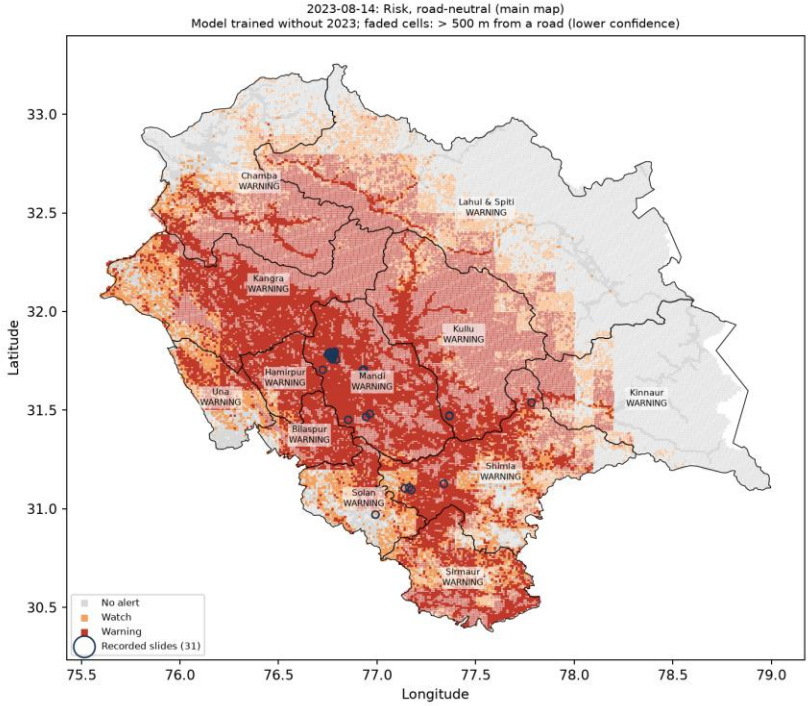
Partly artefact

Source: Our partial-dependence analysis of the live model.

Whole-HP risk map for a real disaster day: 14 Aug 2023

With a model that never saw 2023, all 31 recorded slides fall in alert cells, but two-thirds of the state is flagged





31 of 31

recorded slides in Watch or Warning cells (29 at Warning)

66.8%

of HP's cells at Watch or above (47.2% at Warning)

12 of 12

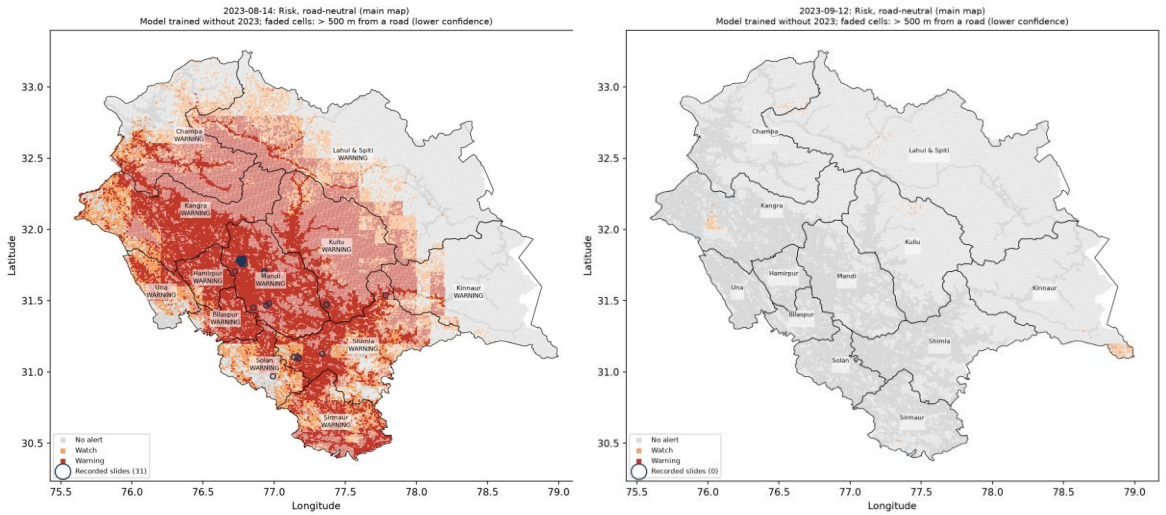
districts at Warning under the provisional 5% rule: too loose

Left: rain on the day (IMERG). Right: risk levels; circles = recorded slides.

Source: Our model run on NASA GPM IMERG V07 rainfall for all HP cells (Google Earth Engine); recorded slides: GSI Bhusanket, NASA Global Landslide Catalog.

Storm day vs quiet day, same model and levels

The system lights up on the disaster day and stays calm on a quiet day; rainy days without slides are the next test



Storm day, 14 Aug 2023
66.8% of HP at Watch or above

Quiet day, 12 Sep 2023
0.4% of HP at Watch or above

Grey: no alert · orange: Watch · red: Warning · circles: recorded slides.

	Storm day 14 Aug 2023	Quiet day 12 Sep 2023
HP mean rain on the day	40.3 mm	0.8 mm
HP at Watch or above	**66.8%**	**0.4%**
HP at Warning	47.2%	0.0%
Districts alerted (of 12)	12	0
Recorded slides in alert cells	31 of 31	none recorded

Caveat, stated plainly

- The quiet day was also nearly dry, so it shows only that the system stays calm without rain
- The harder test, rainy days with no slides, comes from the storm replays (next phase)

Source: Our model runs for 14 Aug and 12 Sep 2023 on NASA GPM IMERG V07 rainfall (Google Earth Engine); recorded slides: GSI Bhusanket, NASA Global Landslide Catalog.

Where we stand against the original goal

Coverage and honesty are on track; accuracy is moderate and limited by the data; automation is the main work left

Original goal	Current status	Verdict
Whole of Himachal Pradesh	All 55,253 cells of 1 km scored; cells > 500 m from a road marked 'lower confidence'	**On track**
As accurate as the data allow	Live model AUC 0.775 (strict test); two-part 0.813. Rain timing capped by satellite rain; records biased to roads	**Moderate; limits measured and explained**
Daily warning to authorities	Daily scoring tested on two real days; district rule provisional; automation and email not built yet	**Main remaining work**
Research questions	RQ1 answered for past rain; RQ2 and RQ3 not started	**Partly**
Honest evaluation	Whole years hidden; rain-rule baseline; every experiment logged, including dropped and wrong ones	**Achieved**
Free tools only	Google Earth Engine, open data, Python	**Achieved**

Source: Results on slides 12–22 of this deck.

Limitations

We state each limit together with what it means for using the system

Records follow roads
Slides are recorded where they hit roads (median 22 m). Skill is measured mostly near roads; remote skill rests on 70 slides (9 in HP).

Timing capped by rain data
Satellite rain pixels are about 10 km; cloudbursts can be smaller. Rain alone times slides as well as our models.

Many slides missed
About 45% of HP slides at Watch and two-thirds at Warning (published nowcasts: 8–60% detected).

False alarms
Live model: about 1 in 4 safe days at a landslide spot reaches Watch; storm days flag large areas.

Small, clustered sample
195 HP slides; 242 of 981 events on one day (2013). HP numbers can move by several points.

Unrecorded slides as negatives
'No slide recorded' is not 'no slide happened'; buffers reduce but cannot remove this.

Sampling artefacts
30-day rain and gentle slopes are distorted by the negative design; to revisit.

Live data
The live model is weaker than the full model; forecast rain adds errors not yet measured.

Source: Our analyses on slides 10–22; Kirschbaum and Stanley 2018 (LHASA detection range).

Mistakes we found and corrected

Our own checks caught 14 mistakes (the main ones below); each was fixed and recorded, including one in Presentation 2

Mistake	How we found it	Correction
Presentation 2 said the 878 background points were inside HP with a 500 m buffer	Mapping the actual file	They spread over three states and beyond; stated on slides 6 and 10
Date coverage inflated (any text counted as a date)	Re-audit with a day-level parser	HP dated records: 10.4% → 1.7%
Some event dates wrong; 'trigger-consistency' claim built on them	October inventory audit	981 cleaned events; claim dropped
Random negatives too easy; score inflated (0.895)	Instructor feedback; strict-vs-easy test	Hard negatives; whole-year testing
Curvature signs wrong (Earth Engine kernel orientation)	Tests on artificial surfaces	Kernels corrected
Road-corridor limit shrank the whole-HP goal	Team review, 29 Oct	Reversed; remote cells marked lower confidence
First RQ1 test unfair (terrain-only score below 0.5)	Result made no physical sense	Rerun on storm-day slopes only (slide 17)
255 border cells dropped, then counted as a 13th district	Count checks on the maps	Fixed; levels for 12 real districts only

Why we list them

- Each mistake was caught by a check we built in: re-audits, count checks, and tests against known answers
- Every experiment, kept, dropped or found wrong (70+ entries), is recorded in the project files, so the results here can be traced and repeated

Source: Our own checks during the project; each change was recorded with its evidence.

Next steps to the final report

Turn the tested pipeline into a running service, test it on many storms, and design how its impact would be assessed

1. **Daily automation**
   Automatic morning run, district summary, email to authorities, map app

2. **Storm replays**
   Several storms and quiet periods, each with a model that never saw that year; set final levels and district rule

3. **Better training data**
   Decide on regenerating negatives (v2); adopt the two-part model if confirmed; test IMD gridded rain; score the AHP baseline on the same test

4. **Impact assessment**
   Method, survey questions for disaster authorities and road department, alert feedback format, pilot plan

5. **Road-blockage study**
   Optional: what free satellite data can and cannot show after a slide

6. **Final paper**
   Methods, results, limitations, plus the impact-assessment section

Source: Final-report requirements from the instructor's mail.

References

Every method choice above rests on one of these sources or on a data test documented in the project files

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Data: GSI Bhusanket inventory; NASA GPM IMERG V07; CHIRPS (Funk et al. 2015); ERA5-Land; Copernicus DEM; MERIT Hydro; ESA WorldCover; Landsat; GSI NGDR; OpenStreetMap; IMD national bulletins; NASA LHASA README.

Source: All references were checked against the publisher or an index during the project.